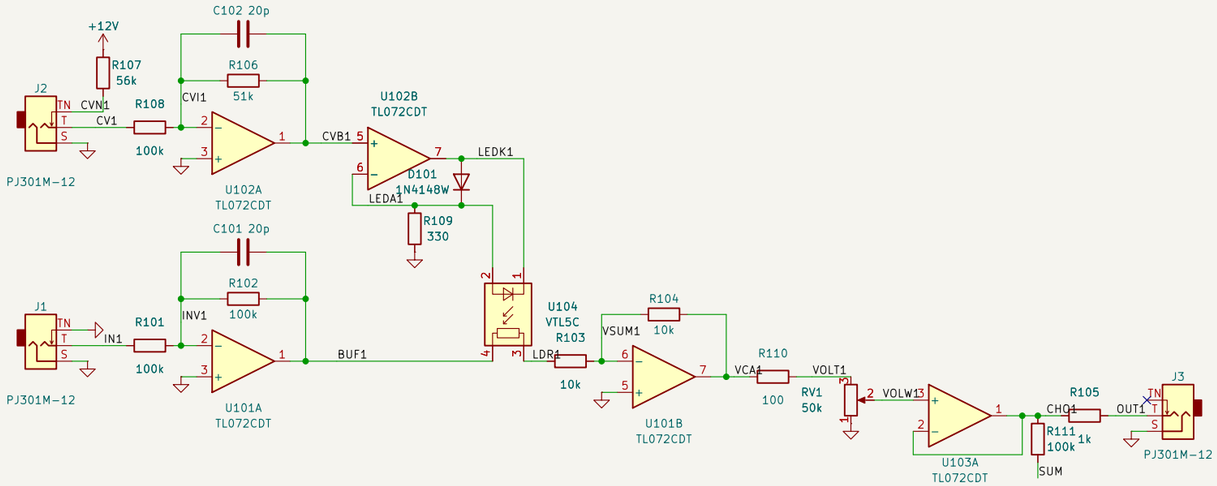
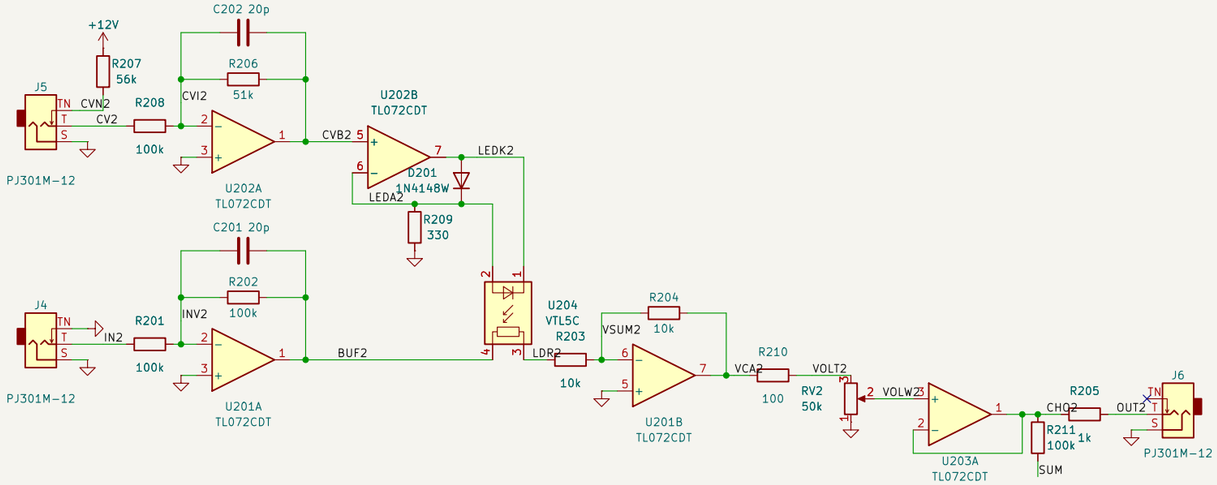
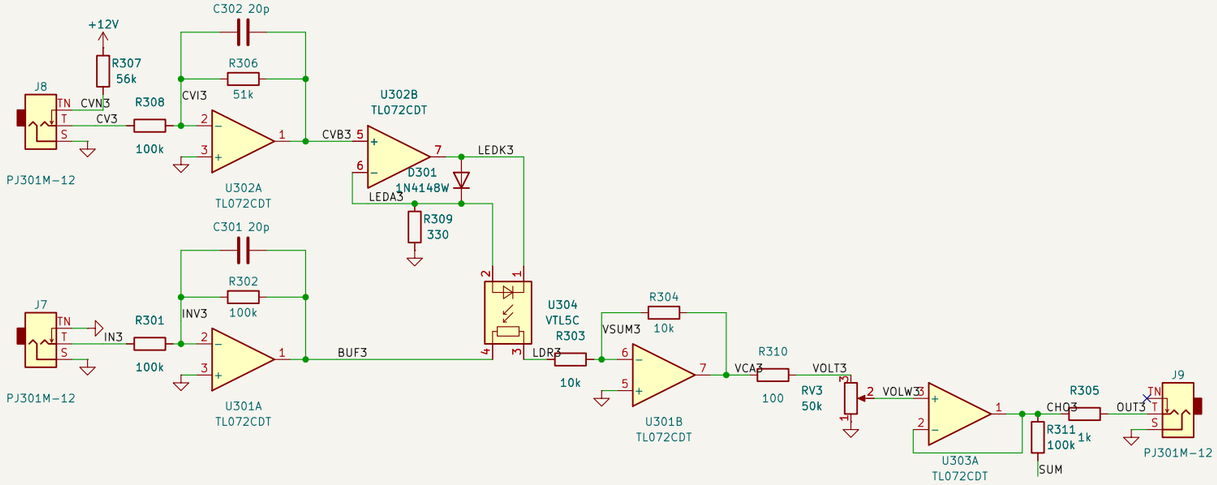
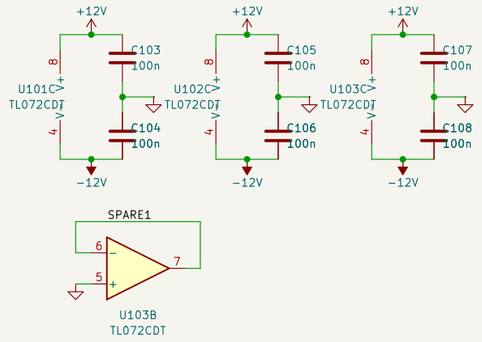
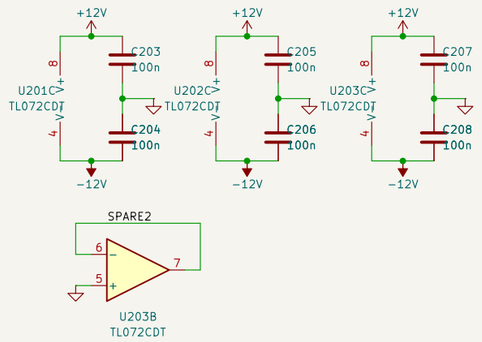
BIG MODULAR VCA–MIX – S4 schematic: 3 x vactrol VCA (Bread Modular V2CA at CURVE 0, LOW/SVF vactrol driver) + post–VCA volume + 3–channel mixer (modules/imix). Circuit intent: docs/research.md

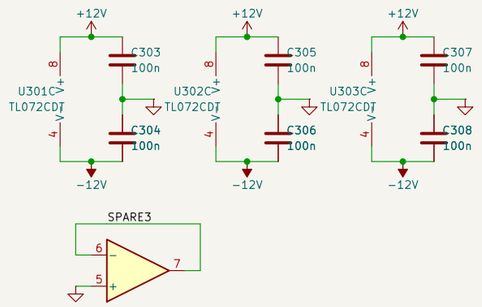


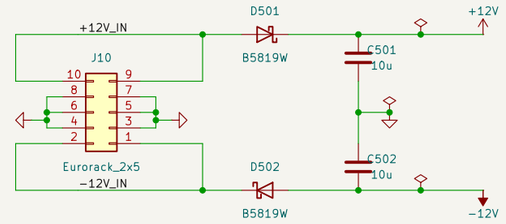








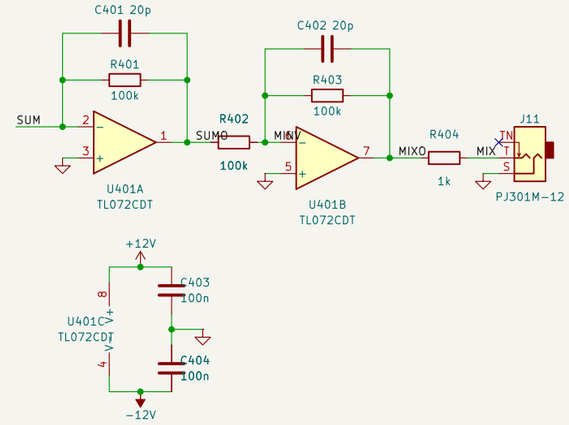




POWER: Eurorack +12V / –12V –> B5819W per rail –> +12V / –12V (about +/–11.6 V). +5V and pins 11–16 unused. 100n at every op–amp supply pin, 10u bulk per rail.

PWR_FLAG: #FLG01 marks the Eurorack GND; #FLG02/#FLG03 mark the +12V/–12V rails, fed from the header only through D501/D502, which ERC cannot see through.

MIX (modules/imix): CHO n –> 100k –> SUM –> summer (100k || 20p) –> inverter –> 1k (2512) –> MIX. MIX = CH01 + CH02 + CH03 with the input polarity. All outputs 1k / 1 W 2512.



CHANNEL n (n = 1..3: refs Rn01..Rn11, Cn01..Cn0B, Dn01, Un01..Un04):
every input buffered at a fixed 100k.
IN –> –1 buffer –> LDR + 10k –> –1 VCA stage (gain 10k / (10k + R_LDR)) –> 100R –> VOL n –> follower –> 1k –> OUT n, and 100k –> SUM (OUT is post–volume).
CV –> 100k –> –0.51 buffer (CVB) –> vactrol driver, LED in the op–amp loop:
I_LED = 0.51 x CV / 330R, linear from 0 V.
No CV plug: TN normals CV to +12V via 56k (~12 mA, VCA open).

Circuit intent: docs/research.md; placement: docs/geometry.json
SCHEMATIC S4 – 3 x vactrol VCA + mixer, buffered inputs/CV, +–12 V, 6HP – not released

Big Modular

Sheet: /
File: vca–mix.kicad_sch

Title: Big Modular – VCA–MIX

Size: A3	Date:	**Rev: S4**
KiCad E.D.A. 10.0.6		Id: 1/1